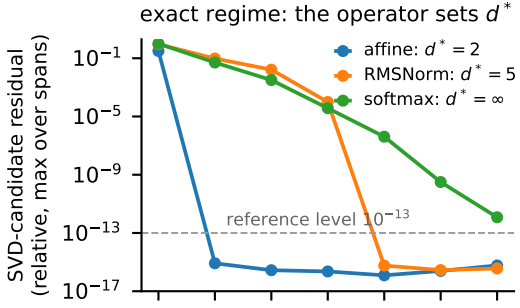
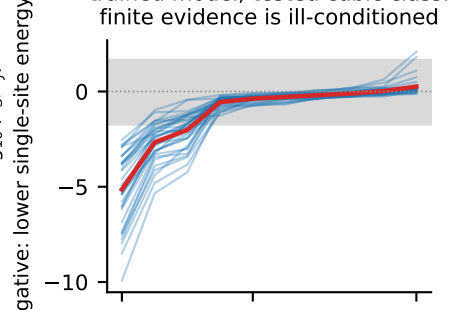

exact regime: the operator sets d^*
affine: $d^* = 2$
RMSNorm: $d^* = 5$
softmax: $d^* = \infty$
reference level 10^{-13}
SVD-candidate residual
(relative, max over spans)
10^{-1}
10^{-5}
10^{-9}
10^{-13}
10^{-17}
1
2
3
4
5
6
7
degree of the mechanism class


trained model, tested cubic class:
finite evidence is ill-conditioned
held-out $\log_{10}(E_S/E_J)$
(negative: lower single-site energy)
0
−5
−10
1
5
10
rank of held-out value
(ten train-fitted directions)